

BING-YUE WU

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EDUCATION

Arizona State University <i>Doctoral Studies in Electrical Engineering</i> Ph.D. candidacy achieved, 2026	2023 – 2026 Tempe, Arizona
National Taiwan University of Science and Technology (Taiwan Tech, NTUST) <i>M.S. in Electrical Engineering</i>	2021 – 2023 Taipei, Taiwan
National Taiwan University of Science and Technology (Taiwan Tech, NTUST) <i>B.S. with major in Electrical Engineering and minor in Computer Science and Information Engineering</i>	2016 – 2020 Taipei, Taiwan

SKILLS

Programming Languages: C, C++, Python, Tcl
EDA Tools: OpenROAD, OpenSTA, Innovus, Genus, ICC2, Design Compiler, HSpice, Virtuoso, Calibre
AI Tools: Claude Code CLI, Codex CLI, Cursor CLI
Language Ability: Mandarin (Native), English (Fluent)

PUBLICATIONS

- V. A. Chhabria, A. Dey, A. B. Kahng, I. Savidis, P. Shrestha, and **B.-Y. Wu**, “**ResizerAgent: Agentic Strategy Selection for Timing Optimization in OpenROAD Resizer**”, in *Proc. MLCAD*, 2026.
- Z. Jiang, Q. Wu, **B.-Y. Wu**, and J. Zhang, “**CAPO: Certification-Guided Agentic Workflow for Physical Design Parameter Optimization**”, in *Proc. GLSVLSI*, 2026.
- B.-Y. Wu**, C.-T. Ho, H. Yang, C.-C. Chang, A. B. Akkur, B. Khailany, and V. A. Chhabria, “**Special Research Session: Toward Agentic Solution for DRC Challenges in Digital VLSI Design**”, in *Proc. DAC*, 2026. [[GitHub](#)]
- B.-Y. Wu**, A. Dey, A. Rovinski, and V. A. Chhabria, “**Focus Session: Large Language Models in Physical Design: From Data Generation to Intelligent Agents**”, in *Proc. DATE*, 2026.
- B.-Y. Wu** and V. A. Chhabria, “**DALI-PD: Diffusion-based Synthetic Layout Heatmap Generation for ML in Physical Design**”, in *Proc. ASP-DAC*, 2026. [[GitHub](#)]
- V. A. Chhabria, A. Ghose, V. Gopalakrishnan, A. B. Kahng, S. Kundu, Y. Liu, Z. Wang, and **B.-Y. Wu**, “**IEEE DATC RDF-2025: Enabling an EDA Research Ecosystem**”, in *Proc. ICCAD*, 2025.
- B.-Y. Wu**, U. Sharma, A. Rovinski, and V. A. Chhabria, “**OpenROAD Agent: An Intelligent Self-Correcting Script Generator for OpenROAD**”, in *Proc. ICLAD*, 2025. [[GitHub](#)]
- V. A. Chhabria, V. Gopalakrishnan, A. B. Kahng, S. Kundu, Z. Wang, **B.-Y. Wu**, and D. Yoon, “**Strengthening the Foundations of IC Physical Design and ML EDA Research**”, in *Proc. ICCAD*, 2024.
- V. A. Chhabria, **B.-Y. Wu**, U. Sharma, K. Kunal, A. Rovinski, and S. S. Sapatnekar, “**Generative Methods in EDA: Innovations in Dataset Generation and EDA Tool Assistants**”, in *Proc. ICCAD*, 2024.
- B.-Y. Wu**, R. Liang, G. Pradipta, A. Agnesina, H. Ren, and V. A. Chhabria, “**2024 ICCAD CAD Contest Problem C: Scalable Logic Gate Sizing Using ML Techniques and GPU Acceleration**”, in *Proc. ICCAD*, 2024. [[GitHub](#)]
- U. Sharma*, **B.-Y. Wu***, S. R. D. Kankipati, V. A. Chhabria, and A. Rovinski, “**OpenROAD-Assistant: An Open-Source Large Language Model for Physical Design Tasks**”, in *Proc. MLCAD*, 2024. [[GitHub](#)]
- V. Gopalakrishnan, **B.-Y. Wu**, and V. A. Chhabria, “**ML-INSIGHT: Machine Learning for Inrush Current Prediction and Power Switch Network Improvement**”, in *Proc. ISLPED*, 2024.
- B.-Y. Wu**, U. Sharma, S. R. D. Kankipati, A. Yadav, B. K. George, S. R. Guntupalli, A. Rovinski, and V. A. Chhabria, “**EDA Corpus: A Large Language Model Dataset for Enhanced Interaction with OpenROAD**”, in *Proc. LAD*, 2024. (Best Paper Nominated) [[GitHub](#)]

V. A. Chhabria, W. Jiang, A. B. Kahng, R. Liang, H. Ren, S. S. Sapatnekar, and **B.-Y. Wu***, “**OpenROAD and CircuitOps: Infrastructure for ML EDA Research and Education**”, in *Proc. VTS*, 2024. (Primary Author) [[GitHub](#)]

B.-Y. Wu, S.-Y. Fang, H.-W. Chang, and P. Wei, “**SpeedER: A Supervised Encoder-Decoder Driven Engine for Effective Resistance Estimation of Power Delivery Networks**”, in *Proc. MLCAD*, 2022. (Best Paper Award)

WORK EXPERIENCE

Arizona State University <i>Graduate Research Assistant</i>	Aug. 2023 – Present Tempe, Arizona
- Developed a large-scale LLM dataset for physical design tool (OpenROAD) interaction, built a domain-specific LLM to generate OpenROAD scripts and answer OpenROAD-related queries, and created a self-correcting agent that iteratively refines scripts using real tool feedback. - Proposed a diffusion-based generative framework to generate synthetic circuit layout heatmaps and open-sourced 23K+ synthetic samples to address data scarcity in ML-driven physical design research. - Investigated LLM-based automated design rule violation (DRV) fixing and developed benchmarking infrastructure to evaluate LLM-driven DRC methodologies. - Served as Topic Chair for ICCAD 2024 CAD Contest (Problem C) and contributed to OpenROAD/CircuitOps infrastructure to advance ML-EDA research and education.	
Synopsys Inc. <i>Intern (Technical-Engineering)</i>	Oct. 2021 – June 2022 Taipei, Taiwan
- Researched a novel Machine Learning-based (ML) solution estimating the effective resistance of Power Delivery Networks in advanced VLSI designs to speed up runtime and improve the accuracy of ML-driven IR analysis tools. - Responsible for designing the data pipeline, the ML model architecture, and the entire effective resistance estimation workflow.	
Academia Sinica <i>Full-time Research Assistant at Computational Finance and Data Analytics Lab</i>	Jul. 2020 – Nov. 2020 Taipei, Taiwan
- Conducted research on a novel Transformer Encoder-based model with financial number category awareness, designing new pre-training and fine-tuning tasks to enhance its performance. - Developed an Online Loan Application Recommender System for E.SUN Commercial Bank, achieving nearly 300% performance improvement. Responsibilities included workflow design, model development, and building Python APIs for industrial deployment.	

AWARDS

Qualcomm Innovation Fellowship North America (QIF)	May 2026
MLCAD Student Travel Grant	Sept. 2024
Ferdinand A. Stanchi Fellowship	Aug. 2024
DAC Young Fellow Travel Grant	Jun. 2024
MLCAD Student Travel Grant	Sept. 2023
Fulton Fellows Fellowship	Aug. 2023
Best Paper Award at MLCAD 2022	Sept. 2022

PROFESSIONAL SERVICE

Reviewer Activities <i>ACM TODAES (2025×2), IEEE TCAD (2026×3)</i>	
Topic Chair of Problem C at 2024 ICCAD CAD Contest <i>IEEE CEDA</i>	Oct. 2024 Newark, New Jersey
2024 ASP-DAC Tutorial Talk <i>ACM SIGDA</i>	Jan. 2024 Incheon, South Korea